COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
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ACTIVITY NO. 0 **Familiarization for the new normal Lab**

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication

1. The simplest way to generate a key pair is to run ssh-keygen without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user authentication are usually stored in the users .ssh directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case id_rsa when using the default RSA algorithm. It could also be, for example, id_dsa or id_ecdsa.
2. Issue the command ssh-keygen -t rsa -b 4096. The algorithm is selected using the -t option and key size using the -b option.
3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that it cannot be used even if someone obtains the private key file. The passphrase should be cryptographically strong.
4. Verify that you have created the key by issuing the command ls -la .ssh. The command should show the .ssh directory containing a pair of keys. For example, id_rsa.pub and id_rsa.

```
(base) nikko@Nikko:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/nikko/.ssh/id_rsa):
Created directory '/home/nikko/.ssh'.
Enter passphrase for "/home/nikko/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/nikko/.ssh/id_rsa
Your public key has been saved in /home/nikko/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:xN5beqC44/jxf5XjFMm306qrFhSB1Rxi1OaGW4rxk0I nikko@Nikko
The key's randomart image is:
+---[RSA 4096]----+
|         +*=..   |
|       .....=    |
|        o  = . . |
|       oE.o + + .|
|       .S=o*.  +o|
|       .o.B=  =o.|
|      o ..oo.+ o.|
|     ..+  ... o  |
|    .o+..ooooo   |
+----[SHA256]-----+
(base) nikko@Nikko:~$ ls -la .ssh
total 16
drwx------  2 nikko nikko 4096 Jul 27 12:39 .
drwx------ 22 nikko nikko 4096 Jul 27 12:38 ..
-rw-------  1 nikko nikko 3381 Jul 27 12:39 id_rsa
-rw-r--r--  1 nikko nikko  737 Jul 27 12:39 id_rsa.pub
```

Task 2: Copying the Public Key to the remote servers

1. To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh-copy-id tool.
2. Issue the command similar to this: ssh-copy-id -i ~/.ssh/id_rsa user@host
3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.

```
(base) nikko@Nikko:~$ ssh-copy-id -i ~/.ssh/id_rsa server-1@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/nikko/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
server-1@server1's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/nikko/.ssh/id_rsa 'server-1@server1'"
and check to make sure that only the key(s) you wanted were added.
```

Server 2

```
(base) nikko@Nikko:~$ ssh-copy-id -i ~/.ssh/id_rsa server-2@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/nikko/.ssh/id_rsa.pub"
The authenticity of host 'server2 (192.168.56.103)' can't be established.
ED25519 key fingerprint is SHA256:2Ilm/CEhxwTzY6MhxRJUTkiUBwm2Yksc87LN4XI1sZ0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are alrea
dy installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to instal
l the new keys
server-2@server2's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/nikko/.ssh/id_rsa 'server-2@server2'"
and check to make sure that only the key(s) you wanted were added.

(base) nikko@Nikko:~$
```

Checking:

Server 1;

```
(base) nikko@Nikko:~$ ssh server-1@server1


Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Jul 27 07:15:23 AM UTC 2026

  System load: 0.01               Memory usage: 6%   Processes:       109
  Usage of /:  16.4% of 18.53GB   Swap usage:   0%   Users logged in: 1


Expanded Security Maintenance for Applications is not enabled.

48 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings


Last login: Mon Jul 27 06:42:31 2026 from 192.168.56.1
server-1@Server-1:~$
```

Server 2:

```
(base) nikko@Nikko:~$ ssh server-2@server2
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Jul 27 07:16:53 AM UTC 2026

  System load: 0.11               Memory usage: 7%   Processes:       110
  Usage of /:  12.5% of 24.44GB   Swap usage:   0%   Users logged in: 1


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings


Last login: Mon Jul 27 07:14:59 2026 from 192.168.56.1
server-2@server-2:~$ S
```

4. On the local machine, verify that you can SSH with Server 1 and Server 2. What did you notice? Did the connection ask for a password? If not, why?

- It didn't ask me for my password. Because instead of asking for the password it just scans the public password which is inside of the id_rsa.pub file.

Reflections:
Answer the following:
1. How will you describe the ssh-program? What does it do?
   - The ssh program is for accessing another device remotely. By using the command ssh-copy-id to copy the public key.
2. How do you know that you already installed the public key to the remote servers
   - Try to login your server using ssh.

Task 3: Set up the Git Repository
1. On the local machine, verify the version of your git using the command which git. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: sudo apt install git
2. After the installation, issue the command which git again. The directory of git is usually installed in this location: user/bin/git.
3. The version of git installed in your device is the latest. Try issuing the command git --version to know the version installed.
4. Using the browser in the local machine, go to www.github.com.
5. Sign up in case you don't have an account yet. Otherwise, login to your GitHub account.
a. Create a new repository and name it as CPE232_yourname. Check Add a README file and click Create repository.
b. Create a new SSH key on GitHub. Go your profile's setting and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create a new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.
c. On the local machine's terminal, issue the command cat .ssh/id_rsa.pub and copy the public key. Paste it on the GitHub key and press Add SSH key.
d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the Code drop down menu. Select SSH and copy the link.

e. Issue the command git clone followed by the copied link. For example, git clone git@github.com:jvtaylar-cpe/CPE232_yourname.git. When prompted to continue connecting, type yes and press enter.
f. To verify that you have cloned the GitHub repository, issue the command ls. Observe that you have the CPE232_yourname in the list of your directories. Use CD command to go to that directory and LS command to see the file README.md.
g. Use the following commands to personalize your git.
• git config --global user.name "Your Name"
• git config --global user.email yourname@email.com
• Verify that you have personalized the config file using the command cat ~/.gitconfig

h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.
i. Use the git status command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven't, and which files aren't being tracked by Git. Status output does not show any information regarding the committed project history. What is the result of issuing this command?
j. Use the command git add README.md to add the file into the staging area.
k. Use the git commit -m "your message" to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next commit.
l. Use the command git push <remote><branch> to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository. As an example, you may issue git push origin main.
m. On the GitHub repository, verify that the changes have been made to README.md by refreshing the page. Describe the README.md file. You can notice the how long was the last commit. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote

```
(base) nikko@Nikko:~/CPE212_Andoy-$ git config --global user.name "Francis Nikko I. Andoy"
(base) nikko@Nikko:~/CPE212_Andoy-$ git config --global user.email "nikkandoy9
> ^C
(base) nikko@Nikko:~/CPE212_Andoy-$ git config --global user.email "nikkandoy9@gmail.com"
(base) nikko@Nikko:~/CPE212_Andoy-$ cat ~/.gitconfig
[user]
        email = nikkandoy9@gmail.com
        name = Francis Nikko I. Andoy
```

```
(base) nikko@Nikko:~/CPE212_Andoy-$ which nvim
/snap/bin/nvim
(base) nikko@Nikko:~/CPE212_Andoy-$ /snap/bin/nvim README.md
(base) nikko@Nikko:~/CPE212_Andoy-$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
(base) nikko@Nikko:~/CPE212_Andoy-$ git add README.md
(base) nikko@Nikko:~/CPE212_Andoy-$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

(base) nikko@Nikko:~/CPE212_Andoy-$ git commit -m "Updated README type shi-"
[main 2cc7ec0] Updated README type shi-
 1 file changed, 6 insertions(+), 1 deletion(-)
```

```
(base) nikko@Nikko:~/CPE212_Andoy-$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 375 bytes | 125.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:illuminado22/CPE212_Andoy-.git
   2ed4ffc..2cc7ec0  main -> main
```

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:
Answer the following:
3. What sort of things have we so far done to the remote servers using ansible
Commands?

- I think the commands that were used when connecting to a remote server using ssh. Like for example when connecting to server 1 and 2 using a public key.

4. How important is the inventory file?

- Inventory file that has a configuration inside it really is important, an example of it is when configuring the file where the hosts and the ip addresses are stored in or where the key pairs are stored.

Conclusions/Learnings:

- In conclusion, I've learned how to use a public key to remotely connect to another server using ssh. An example of this is when I connect my local machine to server 1 and 2 using a public key, I also discovered that I can access my repository in github using ssh. I now understand why this kind of technique exists, since the internet is not secured, our data is just flying around, plain and human readable. By using ssh we are assured that our plain text is mathematically encrypted before transmission, ensuring that our communications and code repositories remain completely secure.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini, grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction, idea generation, explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Give my idea and my own understanding of the concepts to check if its correct.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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